Lakshay Gupta

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Education

University of Toronto — Mississauga, ON
Bachelor of Science, Specialist in Computer Science; GPA: 3.8/4.0 — *Sept. 2024 – June 2028*

- **Relevant Coursework:** Software Design, Data Structures & Analysis, Introduction to Databases, Computer Organization, Introduction to Machine Learning, Software Tools and System Programming.

Technical Skills

Languages: Java, Python, C, SQL, HTML/CSS, JavaScript, R, MIPS Assembly
Tools & Technologies: Git, GitHub, Linux/Unix, JUnit, Supabase, PostgreSQL, MySQL, SQLite, REST APIs, JSON, Jupyter, Figma, WordPress
Concepts: Object-Oriented Programming, Data Structures & Algorithms, Clean Architecture, SOLID, Machine Learning, Decision Trees, Random Forests, Unix Processes, Pipes, File Descriptors, Testing, Debugging

Projects

LakshayEats – Food Journal and Recipe Finder | *Java, Supabase, PostgreSQL, Spoonacular API, JUnit* — 2025

- Collaborated in a 4-member team to build a Java recipe and food journal application using Clean Architecture, SOLID principles, and GitHub-based version control.
- Integrated Spoonacular REST API with Supabase/PostgreSQL to support recipe search, saved recipes, persistent user preferences, and tested use-case logic with JUnit.

C Wine Classifier with Process-Based Bagging | *C, Unix, fork, pipe, Decision Trees* — 2026

- Built a process-based machine-learning system in C to classify wine samples using decision-tree learners and bagging-style ensemble execution.
- Used `fork()`, Unix pipes, and file descriptors to coordinate parent-child processes, exchange classifier results, and aggregate model outputs.

First-World Country Decision Tree & Random Forest Classifier | *Python, pandas, scikit-learn, Gradio* — 2025

- Developed a machine-learning pipeline to classify countries using socio-economic indicators, including data cleaning, feature engineering, model evaluation, and reproducible demos.
- Implemented decision-tree and random-forest logic, benchmarked against scikit-learn baselines, and achieved approximately 98% test accuracy.

Library Management System | *Python, SQL, SQLite/MySQL, CLI* — 2024

- Engineered a CLI-based library system with role-based access, authentication, inventory tracking, borrower records, transaction history, and CRUD workflows.
- Designed relational schemas and SQL queries for persistent storage, validation, efficient retrieval, and maintainable data operations.

ReturnX – AI-Powered Returns Platform | *Hack the Future 2025, Gemini API, Figma* — 2025

- Designed Figma mockups, product workflows, and pitch materials for an AI-powered e-commerce returns platform focused on product verification and return approvals.
- Contributed to product strategy and presentation development, helping the team place 4th out of 34 teams at Hack the Future 2025.

Additional Projects | *MIPS Assembly, Python, PyGame, JSON* — 2024 – 2025

- Implemented a falling-block Columns game in MIPS Assembly with menu flow, gameplay loop, input controls, difficulty modes, gravity timing, and match detection.
- Built Python projects including a PyGame Pong recreation with OOP-based game logic and a JSON-driven text adventure game with branching paths and inventory logic.

Experience

Tech Support Team Founder & Developer — 2022 – 2024
Lincoln M. Alexander Secondary School — *Mississauga, ON*

- Led a 5-member student team to build and maintain digital guidance resources serving 900+ students, improving access to academic and support information.
- Developed and updated a responsive guidance website using HTML, WordPress/Wix, and feedback from staff and students to improve usability and accessibility.
- Trained peers on website maintenance, troubleshooting, documentation, and basic version-control practices to support long-term upkeep.